

The Link layer and Local area networks:

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The Link layer and Local area networks:

- **Data Link and Network Layer**
- **Error detection and correction,**
- **framing,**
- **MAC protocols,**
- **routing algorithms,**
- **RIP, OSPF, BGP,**
- **subnetting,**
- **CIDR,**
- **VLANs,**
- **switching concepts**

Introduction to link layer services

- Data Link Layer (DLL) is the 4th layer from the top in the OSI model.
- Its main purpose is to provide an interface to the Network Layer which is the 3rd layer in the 7- layer OSI model.
- The main purpose of DLL is to transfer the datagrams from the source machine and hand it over to the network layer destination machine across the link (basic communication between the adjacent nodes) with the least probability of error.

Error Detection

- Noise and electromagnetic interference can corrupt data during transmission, turning a 0 into a 1 or vice versa.
- **Error detection** uses redundant bits appended to the frame to check if the data has been tampered with or corrupted
- **Common Techniques:**
- **Parity Check:** Adds a single "parity bit" to make the total number of 1s either even (Even Parity) or odd (Odd Parity). Useful for catching single-bit errors, but fails if an even number of bits are flipped.
- **Checksum:** The data is broken into segments and added together. The 1's complement of this sum is calculated and sent as the checksum. The receiver performs the same math, and if the result is all zero, the data is accepted.
- **Cyclic Redundancy Check (CRC):** A highly reliable mathematical technique based on binary division. The sender and receiver share a "generator polynomial." The sender appends remainder bits to the data so it is exactly divisible by

Checksum

- Remember these **3 steps**:
- **Add** all data words.
- **Add the carry** back to the sum (end-around carry).
- **Invert all bits** (1's complement) → this is the **checksum**.
- At the receiver:
- Add **data + checksum**.
- If the result is **11111111**, the data is **correct**.
- Otherwise, an **error is detected**.

CRC...

- **Steps**
- Append (**Generator length – 1**) zeros to the data.
- Divide using **XOR (modulo-2) division**.
- Append the **remainder (CRC)** to the original data.
- At the receiver, divide the received data by the same generator.
- **Remainder = 0** → **No error**
Remainder ≠ 0 → **Error detected**
- **Example**
- **Data:** 110101
- **Generator:** 1011
- Append **3 zeros** → 110101000
- Divide by 1011 → **Remainder = 001**
- **Codeword sent:** 110101001

Parity

- **Parity Bit Steps (Short)**
- **Sender Side**
- Count the number of **1s** in the data.
- Choose **Even Parity** or **Odd Parity**.
- Add the parity bit.
- Send the **data + parity bit**.
- **Receiver Side**
- Receive the **data + parity bit**.
- Count the total number of **1s**.
- Check the parity:
 - **Even parity:** Total 1s should be **even**.
 - **Odd parity:** Total 1s should be **odd**.
- If the parity is correct → **No error**.
- If the parity is incorrect → **Error detected**.

Error Correction

- When data is transmitted over a network, **noise, interference, or signal loss** may corrupt some bits. After detecting an error, the receiver must recover the original data. This process is called **Error Correction**.
- There are two primary approaches:
- **Forward Error Correction (FEC):** The receiver **detects *and* fixes the error on its own**. The sender includes extra redundant bits (often using **Hamming Codes**) that allow the receiver to mathematically pinpoint the exact corrupted bit and flip it back.
- **Backward Error Correction (Automatic Repeat Request - ARQ):** The receiver detects the error and requests that the sender retransmit the corrupted frame. It is the standard approach for networks like Ethernet and TCP/IP

Framing

Framing is the process of **dividing a continuous stream of bits into frames.**

- It is performed at the **Data Link Layer (Layer 2)** of the OSI model.
- It enables **reliable transmission, synchronization, and proper processing** of data at the receiver.
- Each frame contains **header, payload (data), and trailer (CRC/FCS).**
- **Purpose of Framing (Short)**
 - Defines clear **frame boundaries.**
 - Carries **source and destination addresses.**
 - Supports **error detection** for reliable data transfer.
 - Ensures **organized and reliable communication.**



Types of Framing

- **1. Fixed-Size Framing**

- All frames have the **same fixed size**.
- No start/end delimiters are needed.
- **Drawback:** Wastes space if data is smaller (internal fragmentation).
- **Solution:** Padding is added to fill unused space.

- **2. Variable-Size Framing**

- Frames have **different sizes**.
- Frame boundaries are identified using:
 - **Length Field:** Specifies frame size.
 - **End Delimiter (ED):** Marks the end of the frame.
- **Advantage:** Better bandwidth utilization.
- **Disadvantage:** Corrupted length field or delimiter can cause framing errors.

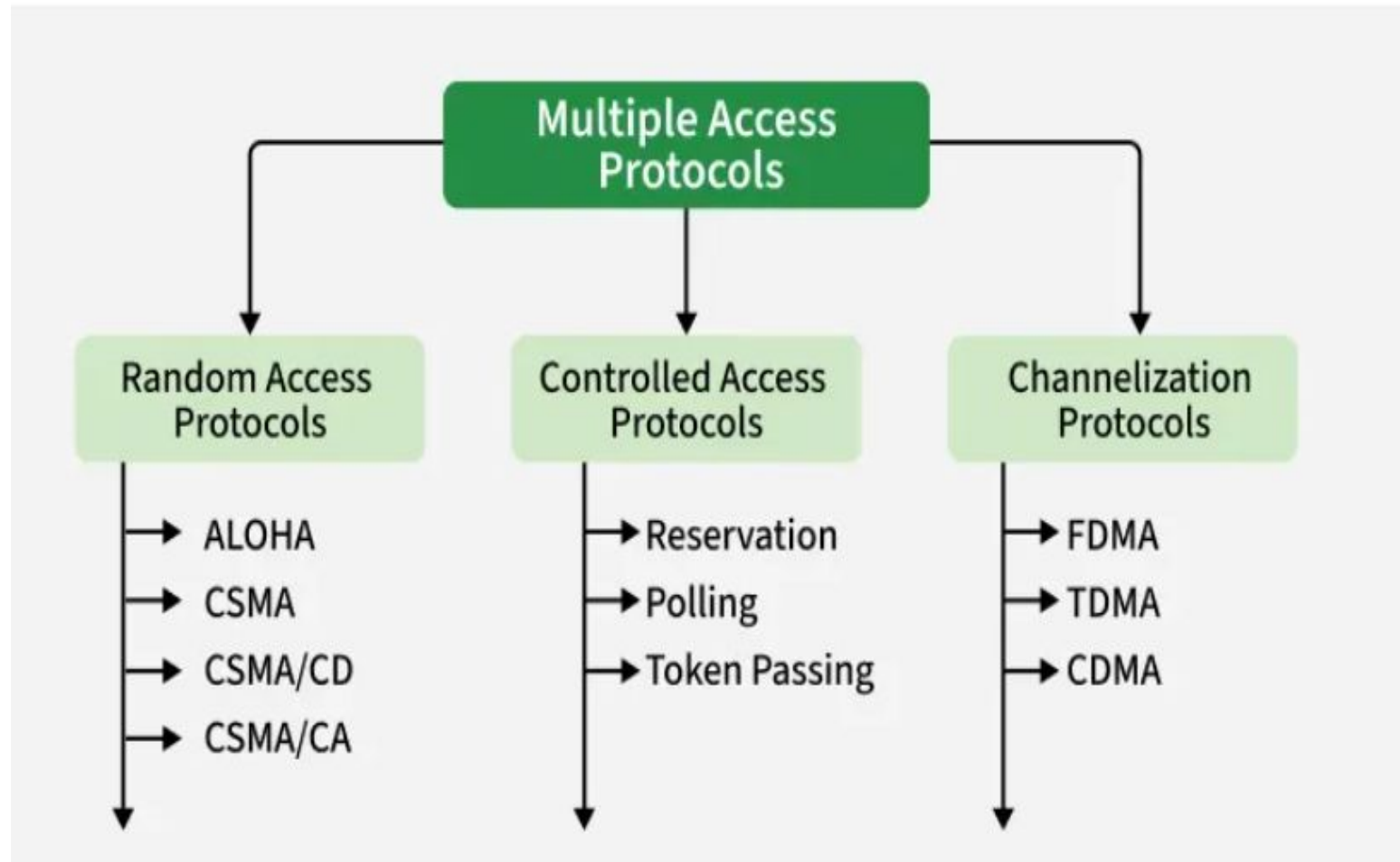
Error detection v/s error correction

Error Detection	Error Correction
The purpose of error detection is to identify the presence of errors	The purpose of error correction is to correct the errors without retransmission
It is generally more efficient (lower overhead)	This can introduce higher overhead and complexity
It is much simpler to implement	It is more complex due to additional coding schemes
It has lower latency (only requires checking)	It contains higher latency (requires decoding and correction)
The error detection is used in networking (e.g., TCP)	The error correction is used in storage systems, error-prone environments (e.g., CDs, DVDs)
Examples of Error detection are 1D/2D Parity Check, CRC, Checksum	Examples of Error correction are Hamming Code, Reed-Solomon, Turbo Codes
This cannot fix errors, only detects them	It is limited to specific types and numbers of errors
It ensures data integrity during transmission	It ensures reliable data retrieval and storage

Multiple access protocols

- The data link layer is only effective if there is a dedicated connection between the sender and the receiver.
- In case there isn't any, numerous stations can access the channel simultaneously.
- Therefore, it is necessary to use **multiple access control (MAC) protocols** to reduce collision and prevent crosstalk.
- MAC protocols are classified into three categories, as illustrated below:

Classification



Random Access Protocols

- In **random access protocol**, no station has a higher priority than another station. Instead, all are equal. There is no predetermined time for sending data; it depends on the channel's status. The order of the stations transmitting data is not predefined. It is sub-divided into the following:
- **ALOHA:** In **ALOHA**, a station can transmit whenever data is available. If another station broadcasts at the same time, collision occurs and the packets are lost. ALOHA protocol has two versions: pure ALOHA and slotted ALOHA.
- **CSMA:** In **Carrier Sense Multiple Access (CSMA)** protocol, each node must monitor the carrier for some time before attempting to deliver data packets. Each node on the bus gets an equal chance to send the data across the network.
- **CSMA/CD: Carrier Sense Multiple Access with Collision Detection (CSMA/CD)** protocol is used for carrier transmission via collision detection. It is used in early ethernet technology for Local Area Networking (LAN).

Controlled access protocol

- These protocols eliminate collisions entirely by establishing an ordered system where only one authorized device can transmit at any given instant.
- **Reservation:** Time is split into intervals containing a reservation frame and a data frame. Stations must reserve a future time slot inside the reservation frame before they are allowed to transmit.
- **Polling:** Utilizes a master-slave topology. A centralized controller (primary device) queries each individual node (secondary device) one by one to check if it has data ready to transmit.
- **Token Passing:** Nodes are organized into a logical ring. A small frame called a "token" is passed sequentially from node to node. A node can only transmit data when it holds the token.

Channelization protocol

- **3. Channelization Protocols**
- Channelization splits the total capacity of the shared medium into smaller, dedicated sub-channels using frequency, time, or unique codes. This allows multiple devices to transmit simultaneously without interfering with one another.
- **FDMA (Frequency Division Multiple Access):** The available bandwidth is divided into separate frequency bands. Each device gets an exclusive, private frequency band for continuous transmission.
- **TDMA (Time Division Multiple Access):** The entire bandwidth is shared, but transmission time is divided into distinct slots. Each device is assigned a specific slot and must synchronize its clock to only transmit during that period.
- **CDMA (Code Division Multiple Access):** All stations transmit simultaneously over the exact same frequency band. Mathematical isolation is achieved by assigning a unique mathematical code to each device's data, which the receiver decodes separately

ALOHA

- The multiple access protocol ALOHA (Advocates of Linux Open-source Hawaii Association) is used to transmit data over a public network channel.
- *Aloha is a data link layered multiple access protocol that describes the way in which multiple terminals can access without interference or conflict.*
- It was developed at the University of Hawaii in the early 1970s to support communication between the university's computers on different islands using UHF wireless radios.
- ALOHA is a MAC (Medium Access Control) protocol in which every station is equally allowed to transmit data.
- As there is more chance of collision, this Aloha protocol is mainly used in fewer stations.
- In this protocol, any user can send the data in any priority, and if there is no other transmission, the data will be transmitted successfully.

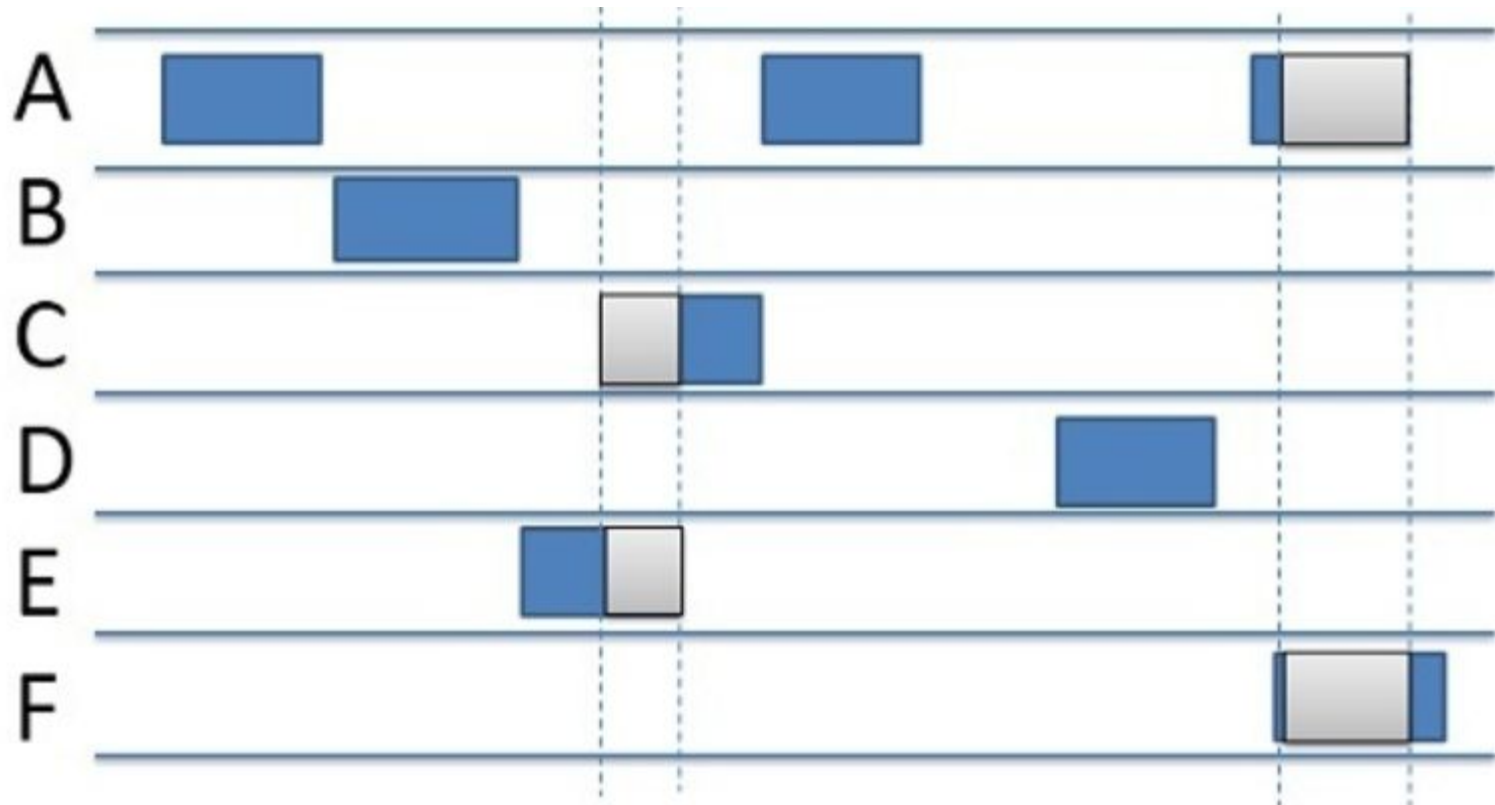
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- **Types of ALOHA**
- Over the years, ALOHA has been developed to improve efficiency and effectiveness. The following are notable versions of ALOHA:

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- **Throughput of Pure ALOHA**
- The probability of successful transmission (S) can be derived from the probability that no other packets are sent during the vulnerable time period. This is given by:
- $S = G \times e^{-2G}$
- where:
- S is the throughput (the average number of successful packet transmissions per packet time).
- G is the average number of packets generated by the system in one packet time .
- **Maximum Throughput of Pure ALOHA**
- The **maximum throughput** occurs when $G=0.5$
- $S_{\max} = 0.5 \times e^{-1} \approx 0.184$
- This means the **maximum throughput of Pure ALOHA** is approximately 18.4%.

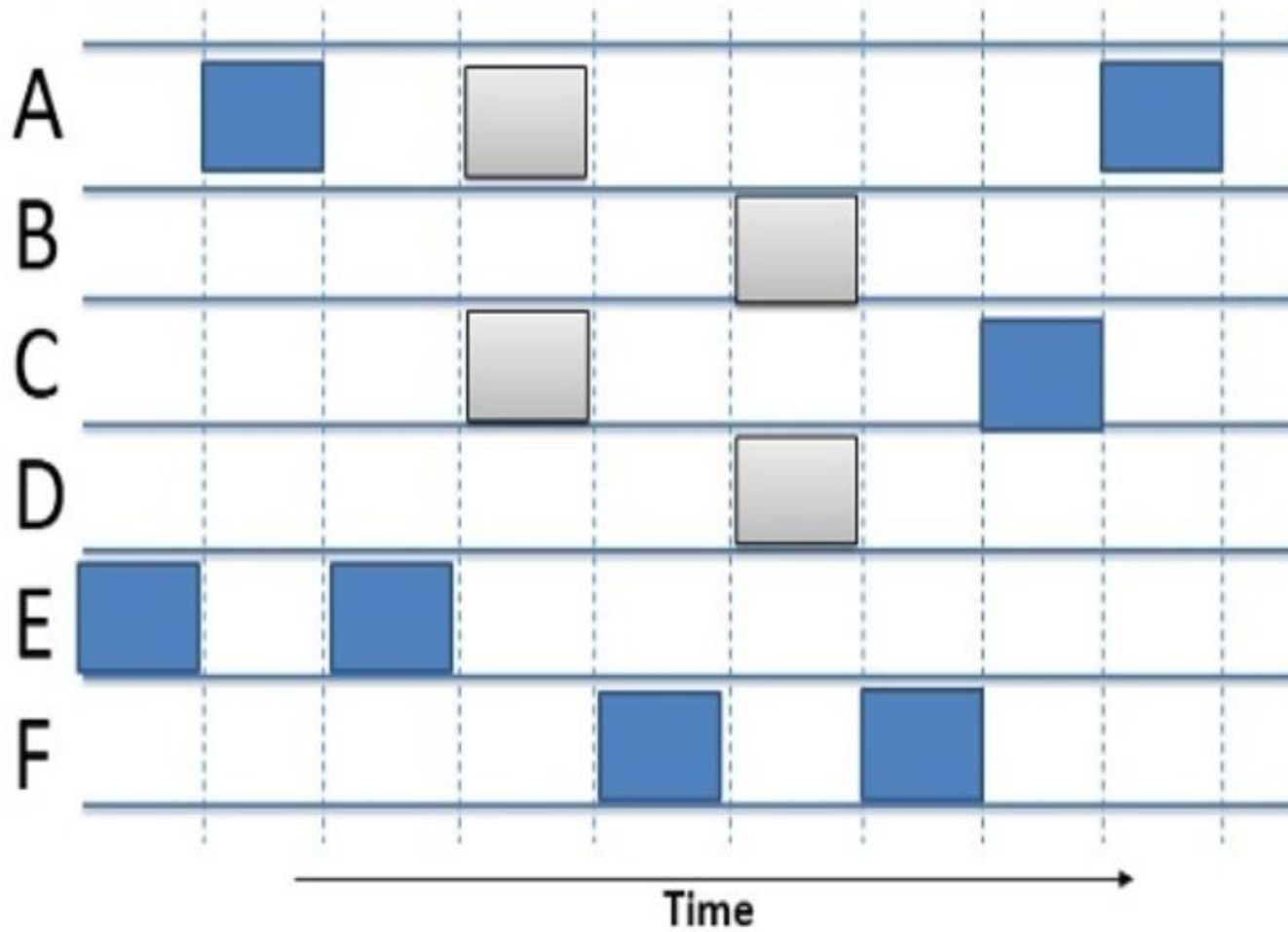
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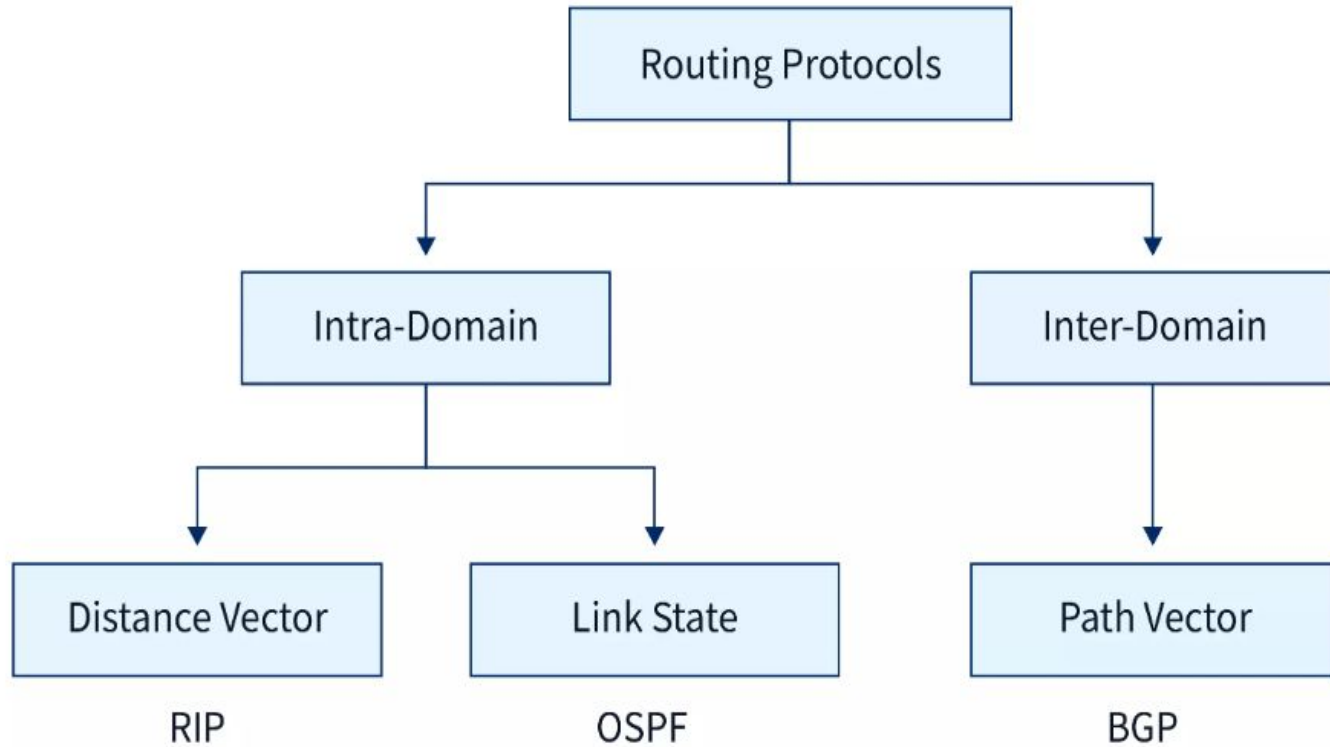
Slotted ALOHA

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA as chances of collision in pure ALOHA are very high.
- In slotted ALOHA, there is still a possibility of collision if two stations try to send at the beginning of the same time slot
- The time in the case of slotted ALOHA is divided into discrete intervals called slots that correspond to a frame.
- Here, slotted ALOHA overcomes the possibility of frame hitting in pure ALOHA.
- Unlike pure ALOHA, this type of ALOHA does not allow data transmission whenever the station wants to send it.
- If a station wants to send the frame to a shared channel, frames can only be sent at the beginning of the slot
- Only one frame is allowed to be sent to every slot. In case the station fails to send data, it will have to wait till the next slot.
- Still, there is a possibility of collision in case two stations try to send a frame at the beginning of the time slot.

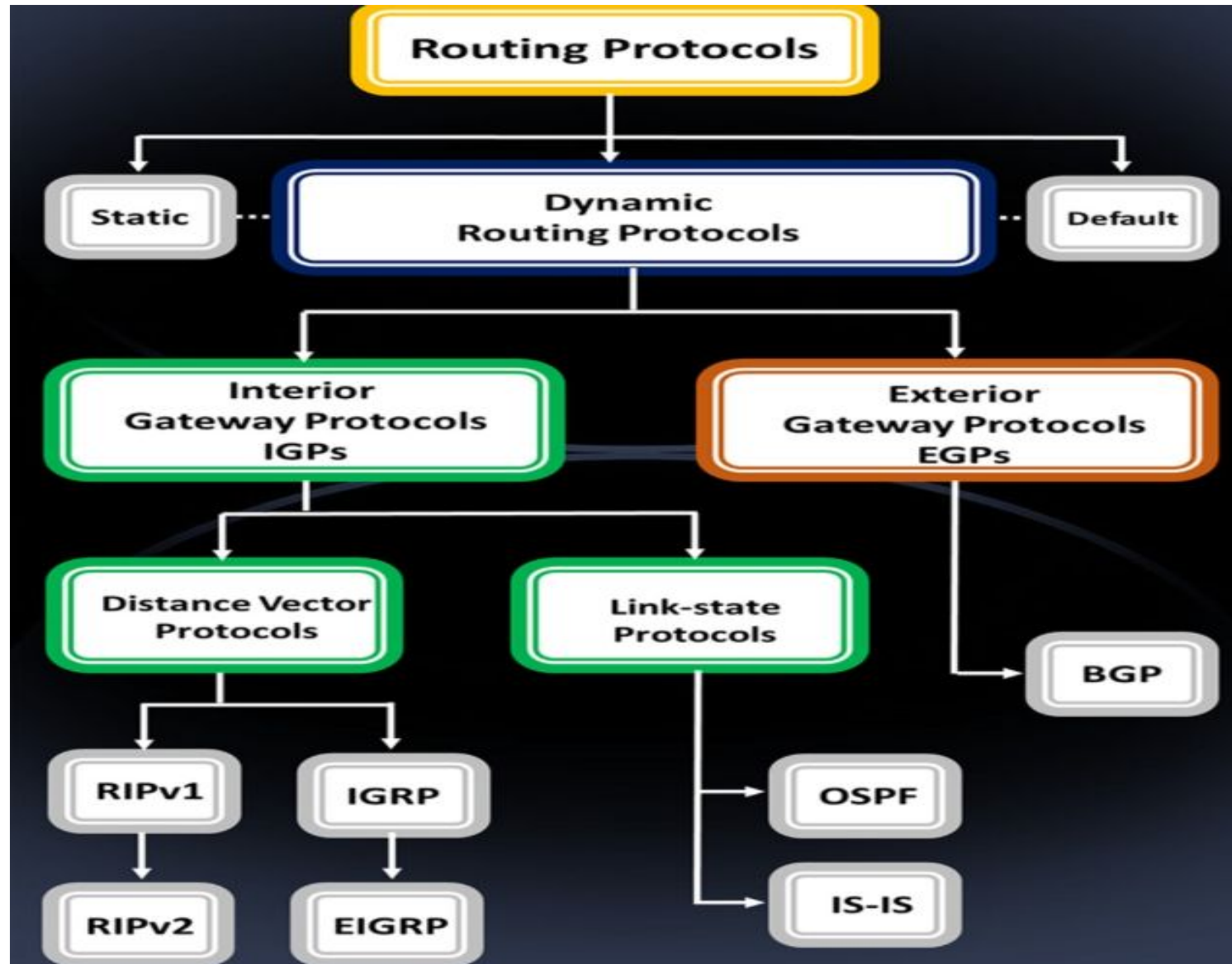
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Routing Protocols..



Routing protocols..



Routing Terminology

1. Autonomous System (AS)

- An **Autonomous System (AS)** is a group of networks that are controlled and managed by **one organization**.
- **Example:** A university network, Reliance Jio network

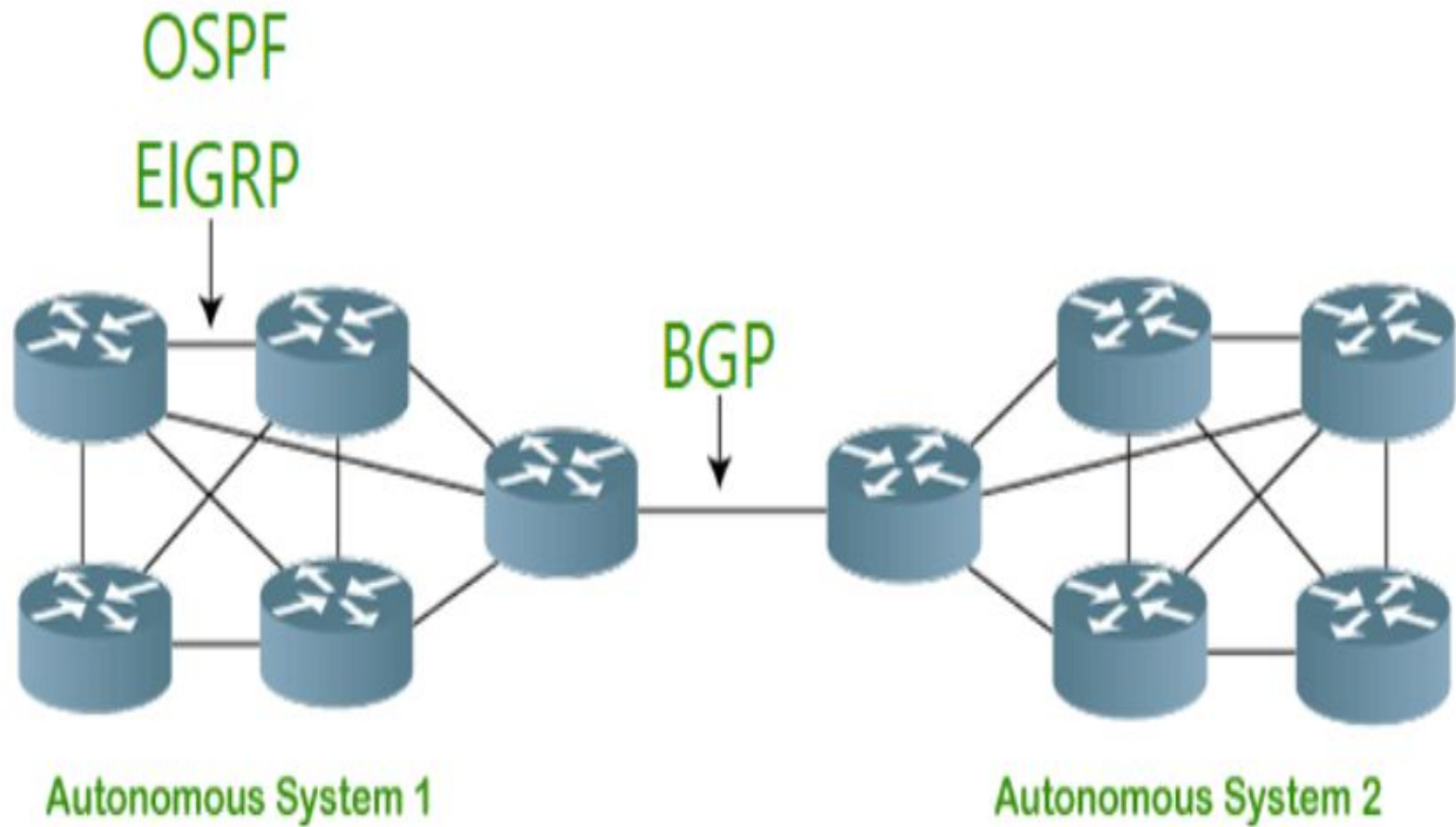
2. Interior Routing Protocol (IRP)

- An **Interior Routing Protocol (IRP)** is used to share routing information **within the same Autonomous System (AS)**.
- **Examples of IRP:** RIP, OSPF, EIGRP

3. Exterior Routing Protocol (ERP)

- **Definition:**
An **Exterior Routing Protocol (ERP)** is used to share routing information **between different Autonomous Systems (ASes)**.
- **Example:**
When Airtel exchanges routing information with Jio, they use **BGP**.

Continue...



Routing Protocols

- Routing algorithms can be broadly categorized into two types, adaptive(static) and nonadaptive(Dynamic) routing algorithms.
- A **Distance Vector** routing protocol is a routing method in which each router shares with its neighboring routers:
- **Distance** = Metric (such as hop count)
- **Vector** = Direction (next hop) to reach a destination
- It is just a **routing technique or concept**.

1. Routing Information Protocol (RIP):

- **Introduction**
- RIP is one of the oldest routing protocols, introduced in the 1980s. It's a **distance-vector protocol**, meaning routers use the number of hops (the number of routers a packet passes through) to determine the best path to a destination.
- **Key Features**
- **Metric:** RIP uses hop count as its primary metric. The maximum number of hops allowed is 15, making RIP unsuitable for large networks.
- **Periodic Updates:** Routers broadcast their entire routing table to neighbors every 30 seconds.
- **Protocol Version:** The two main versions are RIP v1 (classful) and RIP v2 (classless, supporting subnet masks).

Applications/Advantages/Disadvantages

- The Distance Vector Routing Algorithm has several uses:
- **Computer Networking** : It helps route data packets in networks.
- **Telephone Systems** : It's used in some telephone switching systems.
- **Military Applications** : It has been used to route missiles.

What is Link State Routing Algorithm

- A **Link State Routing Protocol** is a routing method in which every router **builds a complete map (topology) of the network** and calculates the shortest path to every destination.
- It is opposite of Distance vector routing algorithm.
- Instead of asking neighbors for the best route, each router knows the network layout and computes the best path itself.
- OSPF is the Example of Link state Routing Algorithm.
- **Characteristics**
 - Every router knows the complete network topology.
 - Uses **Cost** as the routing metric.
 - Uses **Dijkstra's Shortest Path First (SPF) algorithm**.

OSPF

- OSPF (Open Shortest Path First) is a link-state routing protocol designed for routing IP packets **within a single** Autonomous System (AS).
- It dynamically discovers network topology changes and calculates the most efficient paths for data transmission using Dijkstra's Shortest Path First algorithm.
- Unlike distance-vector protocols that rely on hop counts, this protocol evaluates link costs based on bandwidth, creating a more accurate and efficient routing strategy.

Continue...

- **Step 1: Discover Neighbors**
- Each router sends **Hello packets**.
- Routers receiving Hello packets become **neighbors**.
- **Example:**
Router A sends a Hello packet to Router B. Router B replies, and they become neighbors.
- **Step 2: Exchange Routing Information**
- Neighbor routers exchange information about the networks they are connected to.
- Each router builds a **Link-State Database (LSDB)**.
- **Step 3: Calculate the Best Path**
- OSPF uses **Dijkstra's Shortest Path First (SPF) Algorithm**.
- It calculates the **lowest-cost (shortest) path** to each destination.
- **Step 4: Update Routing Table**
- The router stores the best path in its **routing table**.
- Data is then forwarded using this shortest path

Steps of OSPF

- **Neighbor Discovery:** OSPF routers send Hello packets to identify and establish communication with other OSPF-enabled routers on the same link. If their settings match (area ID, hello/dead timers, etc.), they become neighbors.
- **Adjacency Formation:** On multi-access networks (like Ethernet), OSPF elects a Designated Router (DR) and Backup Designated Router (BDR) to reduce traffic overhead. Full adjacencies are formed with the DR and BDR.
- **Link-State Advertisements (LSAs):** Each router generates LSAs containing information about its interfaces, links, and network status. These LSAs are flooded to all OSPF routers within the same area.
- **Link-State Database (LSDB):** All routers build an identical LSDB based on received LSAs. This database reflects the complete network topology from the router's perspective.
- **Shortest Path First (SPF) Algorithm:** Using Dijkstra's algorithm, each router calculates the shortest and most cost-effective paths to every known destination in the network.
- **Routing Table Update:** Once it is complete, the router updates its routing table with the best next-hop decisions.

Example

- (10 Mbps) 10
- R1 ----- R2
- \ /
- 1 \ /1
- (100 Mbps) (100 Mbps)
- \ /
- R3

Solution Using RIP (Distance Vector)

- **Step 1: Calculate Hop Count**
- There are two possible paths.
- **Path 1**
- R1 → R2
- Hop Count = **1**
- **Path 2**
- R1 → R3 → R2
- Hop Count = **2**
- **Step 2: Compare Hop Count**

Path	Hop Count
R1 → R2	1 
R1 → R3 → R2	2

Continue...

- **RIP Decision**
- RIP selects
- $R1 \rightarrow R2$
- **Reason:** RIP always chooses the path with the **minimum hop count.**

Solution Using OSPF (Link State)

- Assign OSPF costs based on bandwidth.

Link	Cost
R1 → R2	10
R1 → R3	2
R3 → R2	2

- **Step 1: Calculate Total Cost**

- **Path 1**

- R1 → R2
- Total Cost = **10**

- **Path 2**

- R1 → R3 → R2
- Total Cost =
- $2 + 2 = 4$

Continue...

- Step 2: Compare Costs

Path	Cost
R1 → R2	10
R1 → R3 → R2	4 

- **OSPF Decision**

- OSPF selects

- R1 → R3 → R2

- **Reason:** OSPF chooses the path with the **lowest total cost**, even if it has more hops.

OSPF Cost calculation

- **OSPF Cost Formula**
- **Reference Bandwidth**
- **= 100 Mbps**

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

- **Example 1**
- Suppose the links are:
- R1 → R2 = **10 Mbps**
- R1 → R3 = **100 Mbps**
- R3 → R2 = **100 Mbps**
- Traditionally, OSPF uses:
- **Reference Bandwidth = 100 Mbps**
- **Example 1**
- Suppose the links are:
- R1 → R2 = **10 Mbps**
- R1 → R3 = **100 Mbps**
- R3 → R2 = **100 Mbps**
- Then,

Continue...

Now calculate the path cost.

- **Path 1**
- $R1 \rightarrow R2$
- Cost = **10**
- **Path 2**
- $R1 \rightarrow R3 \rightarrow R2$
- Cost = **1 + 1 = 2**
- OSPF selects:
- $R1 \rightarrow R3 \rightarrow R2$
- because **2 < 10**.

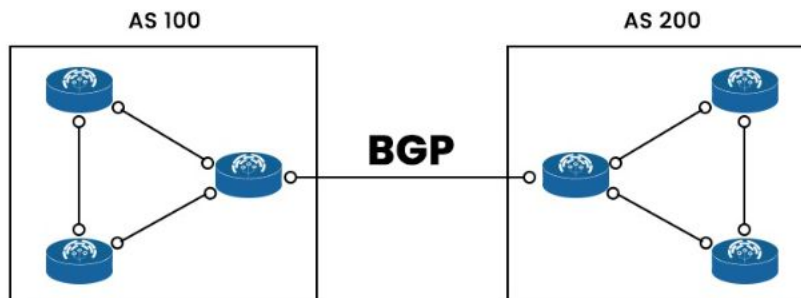
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Link	Bandwidth	Cost
R1 → R2	10 Mbps	$100 \div 10 = \mathbf{10}$
R1 → R3	100 Mbps	$100 \div 100 = \mathbf{1}$
R3 → R2	100 Mbps	$100 \div 100 = \mathbf{1}$

Border gateway protocol

- Border Gateway Protocol (BGP) is an **Exterior Gateway Protocol** (EGP) used to connect autonomous systems (AS) in any topology.
- Each AS must have at least one BGP-enabled router connected to another AS's BGP router.
- Its primary function is to exchange network reachability information between BGP systems.
- It is a **path-vector routing protocol** (unlike distance-vector or link-state protocols).
- Inside an organization, **OSPF** or **RIP** is used to find the best path between internal routers.
- But when data must travel **between different organizations/ISPs**, **BGP** is required.

- When two or more independent systems need to be connected, they use BGP to handle the routing of packets between them. BGP uses the AS to identify a system uniquely.



Types of Border Gateway Protocol

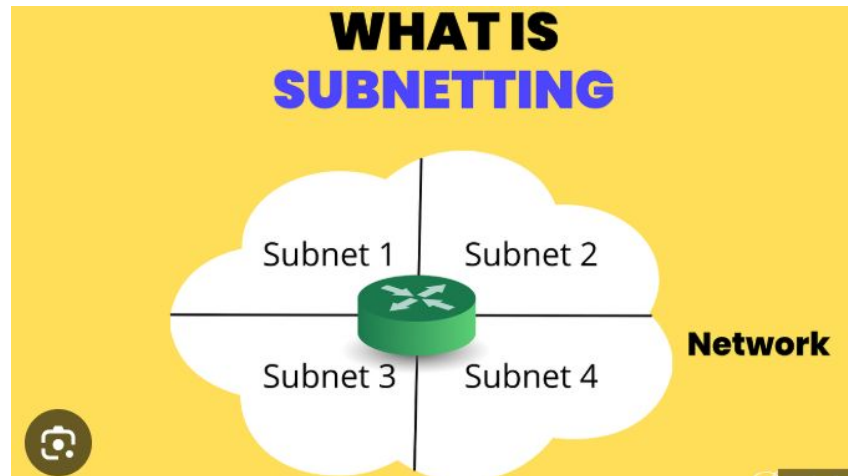
- **1. External BGP**
 - It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP(External Border Gateway Protocol)
- **2. Internal BGP**
 - It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP(Internal Border Gateway Protocol).

Classful Addressing

Address Class	1st Octet range in decimal	1st Octet bits (Blue Dots do not change)	Network (N) and Host (H) Portion	Default mask (Decimal)	Number of possible networks and hosts per network
A	0-127	00000000 - 01111111	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts ($2^{24}-2$)
B	128-191	10000000 - 10111111	N.N.H.H	255.255.0.0	16,384 Nets (2^{14}) 65,534 hosts ($2^{16}-2$)
C	192-223	11000000 - 11011111	N.N.N.H	255.255.255.0	2,091,504 Nets (2^{21}) 254 hosts (2^8-2)

Subnetting

- A subnet is like a smaller group within a large network.
- **Subnetting** is the process of dividing a large network (IP address space) into **smaller, manageable subnetworks (subnets)** by borrowing bits from the host portion of the IP address.
- Subnet makes the network faster and easier to manage and also improves the security of the network. It is done using a **subnet mask** (e.g., 255.255.255.192), which defines how many bits are reserved for the **network** and how many are left for the **hosts**.



Continue...

- **Why subnetting?**
- Reduces network traffic.
- Improves security by separating networks.
- Makes IP address management easier.
- Limits broadcast domains.
- **Example**
- Suppose you have the network:
- **IP Address:** 192.168.1.0/24
- /24 means the subnet mask is **255.255.255.0**
- Total IP addresses = **256** (0–255)
- Usable host addresses = **254** (excluding network and broadcast addresses)
- Now, assume you need **4 subnets**.

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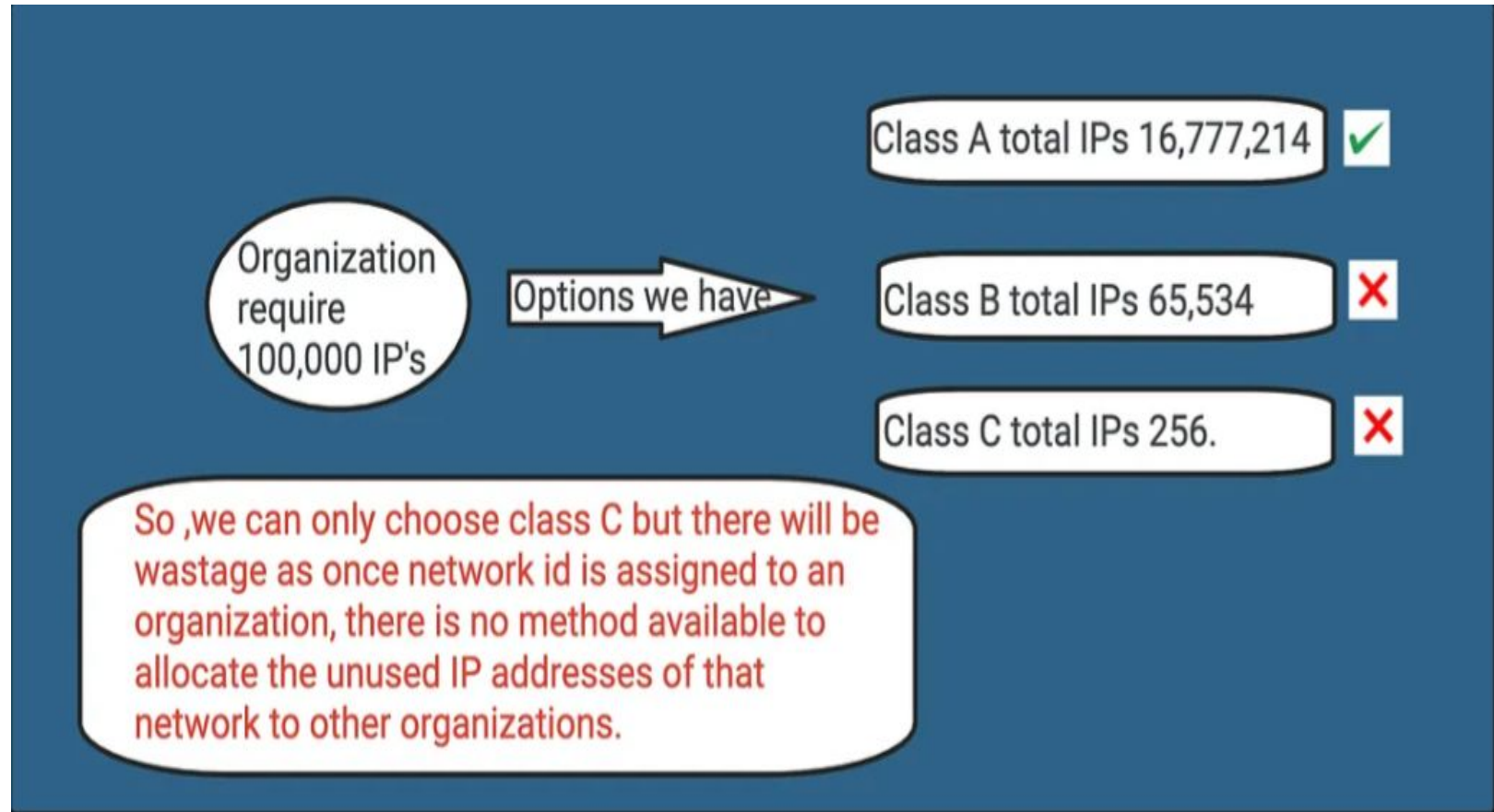
- **Step 1: Borrow bits**
- To create 4 subnets:
- $2^2 = 4$ Borrow **2 bits** from the host portion.
- New subnet mask:
- **/26**
- **255.255.255.192**
- **Step 2: Calculate subnet size**
- Block size:
- $256 - 192 = 64$ Each subnet contains **64 IP addresses**

Continue...

- Step 3: List the subnets

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Need for Classless addressing



Class Inter-Domain Routing(CIDR)

- **Classless addressing**, also called **Classless Inter-Domain Routing (CIDR)**, is an improved IP addressing system.
- It increases the effectiveness of IP address allocation because of the absence of class distribution.
- In 1993 **Class Inter-Domain Routing(CIDR)** is introduced which has Variable Length **Subnet Masks(VLSM)** in which subnet mask is represented as “/X”.
- It has 3 rules
- **Rule 1** – All CIDR block's IP addresses must all be in contiguous state.
- **Rule 2** – The block size must be a power of two to be attractive. Moreover, the block's size is equal to the number of IP addresses in the block.
- **Rule 3** – The block's first IP address must be divisible by the block size

Continue...

- **For example,**
- Assume that the classless address is 192.168.1.35/27
- /27 → First 27 bits are network bits, remaining 6 bits are host bits.
- So, to solve this **classless addressing is introduced in 1993** which solve wastage of ip's and it is more practical

192.168.1.35/27

Total bits(32) - Given range(27) = 5 Hosts bits

Binary representaion = 11000000. 10101000. 00000001. 00100011

Now , Assign 0 to all host i.e, last 5 bits to get first IP of this range

Binary representaion = 11000000. 10101000. 00000001. 00100000

So , first IP is 192.168.1.32

Now , Assign 1 to all host to get last IP of this range

Binary representaion = 11000000. 10101000. 00000001. 00111111

So , Last IP is 192.168.1.63

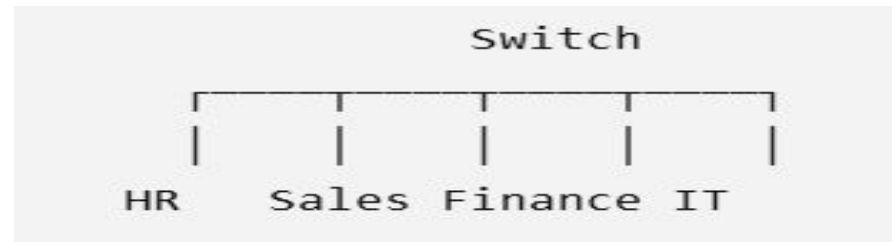
Range = 192.168.1.32 to 192.168.1.63 i.e 31 IPs

VLAN (Virtual Local Area Network)

- **Definition**
- A **VLAN (Virtual Local Area Network)** is a **logical division of a physical LAN**. It allows a **managed switch** to divide one physical network into multiple virtual networks.
- Each VLAN acts as a **separate LAN** even though all devices are connected to the same physical switch.

Example

- Suppose a company has one switch connecting all departments.

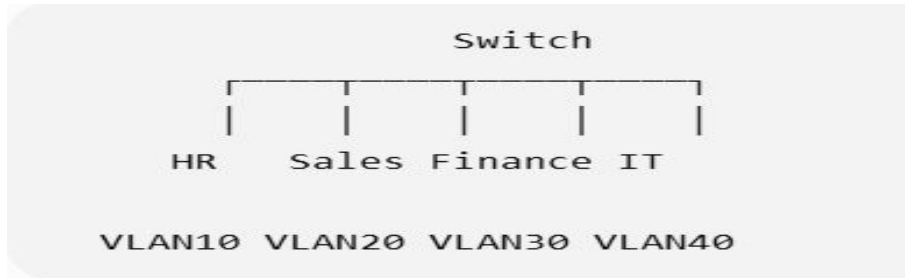


- **Without VLAN:**

- All devices belong to the **same LAN** and receive broadcast messages.

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- With VLAN:



- Each department is placed in a **different VLAN**, so they behave like separate networks even though they use the same switch.
- **Advantages of VLAN**
 - Improves network security.
 - Reduces broadcast traffic.
 - Easier network management.
 - Better network performance.

Continue...

- Allows logical grouping instead of physical grouping.
- **Applications**
- Schools and colleges
- Offices
- Banks
- Universities